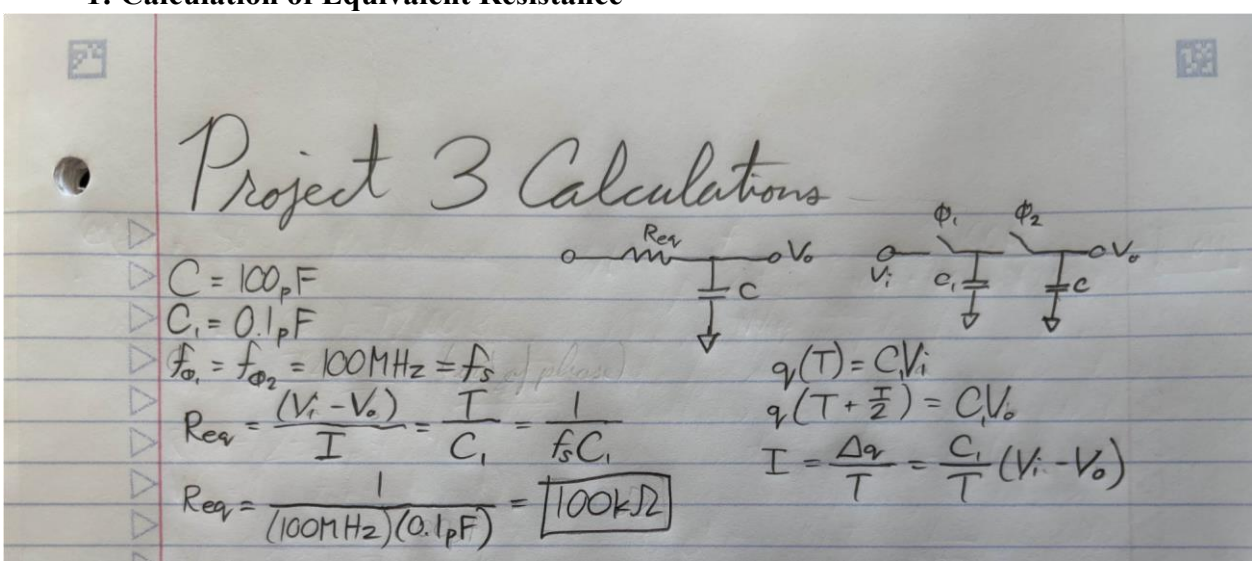
Project #3: S-To-Z Transformations and S-domain Modeling of SC Circuits

1: Calculation of Equivalent Resistance



R_eq = 100k Ohms

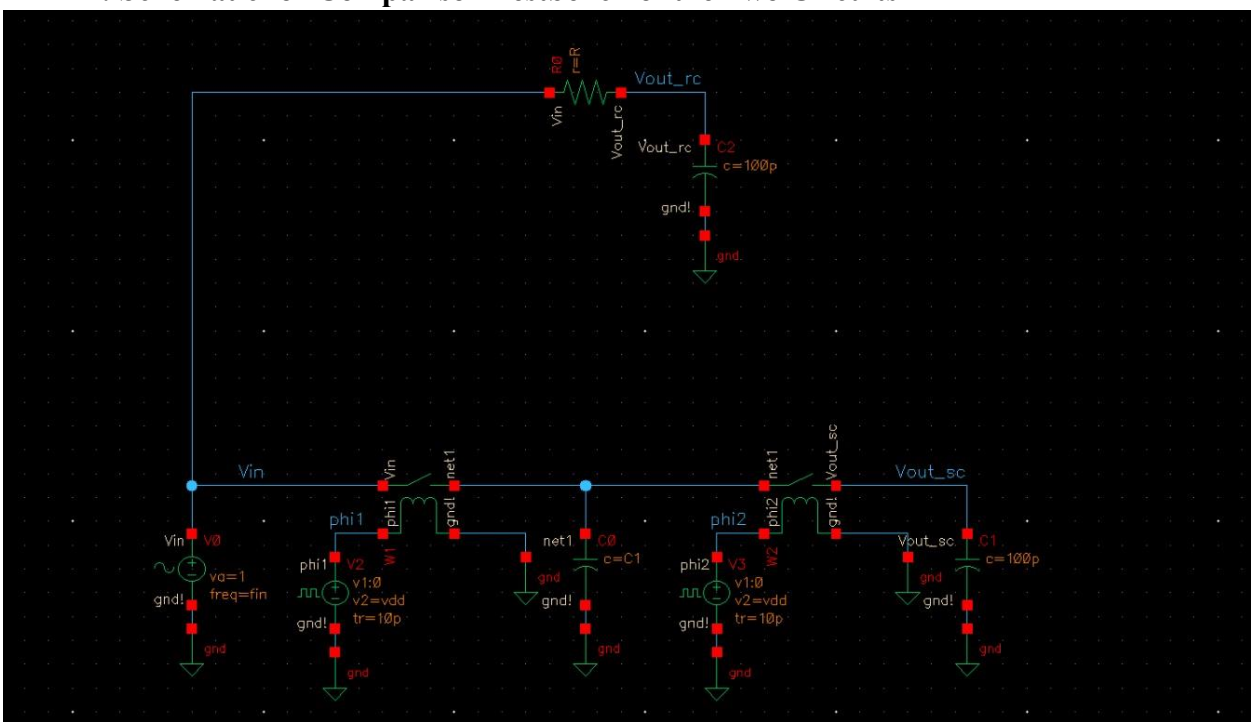
Design Description of Testbench CIrcuit

Shown in Figure 2 is the design of the testbench for the switched capacitor and RC circuits drawn in Figure 1. These circuits both represent a low-pass-filter. The input voltage is wired to both circuits, since they will be tested under the same conditions, and their output voltages will be compared. If the response for both outputs is the same, this proves that the calculated resistance (R_eq = 100k Ohms) makes both circuits equivalent. To achieve the non-overlapping clocks in phi_1 and phi_2, the pulse width for phi 1 is defined as 1/fs/2 (the period of the 100MHz clock), and the pulse width for phi_2 is defined as 1/fs/2 + 1/fs/2*0.05s plus a delay time of the same amount. This offsets both the pulse width and the pulse timing by 5% to ensure that the phi_1 and phi_2 clocks do not overlap. The switches used in the circuit represent ideal

Figure

switches. To minimise simulation wait times, fs is defined as 1000 times the input AC frequency for the 100 Hz and 1kHz transient responses shown in Figure 4 and Figure 5 respectively.

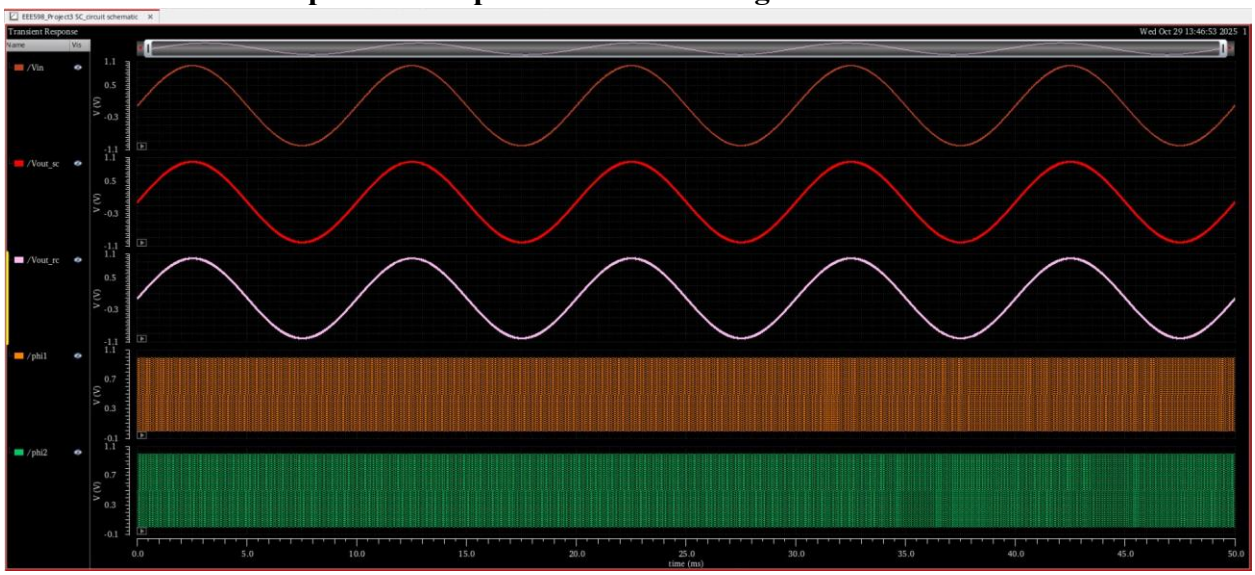
2: Schematic for Comparison Testbench of the Two Circuits



Figure

Figure 3: Transient Response for Input DC Voltage

4: Transient Response for Input 100 Hz AC Voltage



Figure

Figure 5: Transient Response for Input 1 kHz AC Voltage

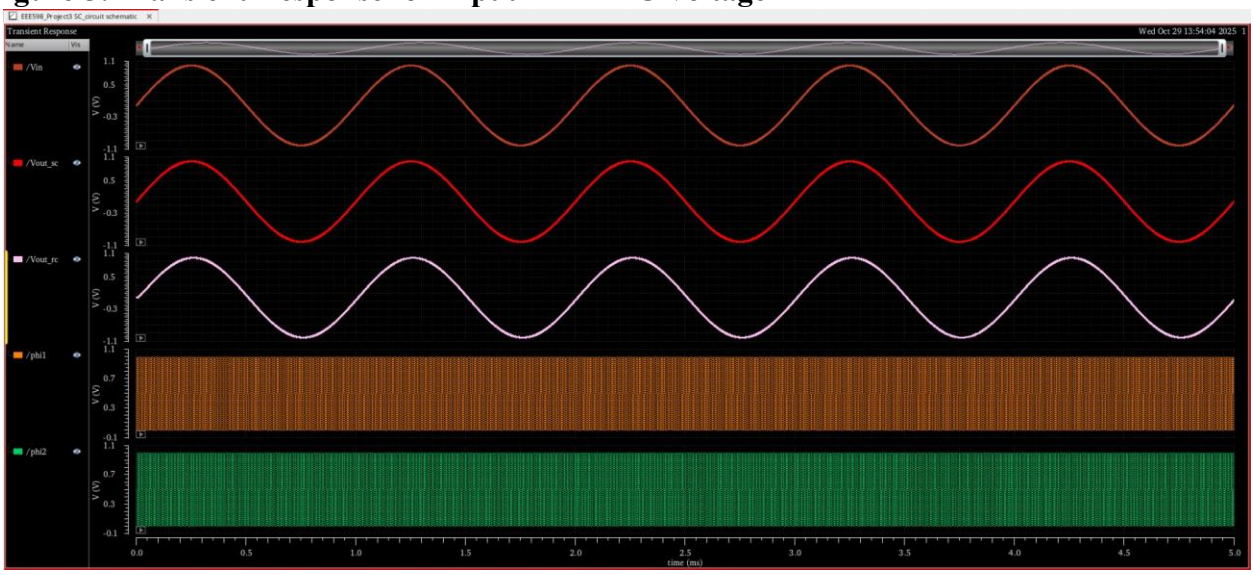


Figure 6: Transient Response for Input 10 kHz AC Voltage

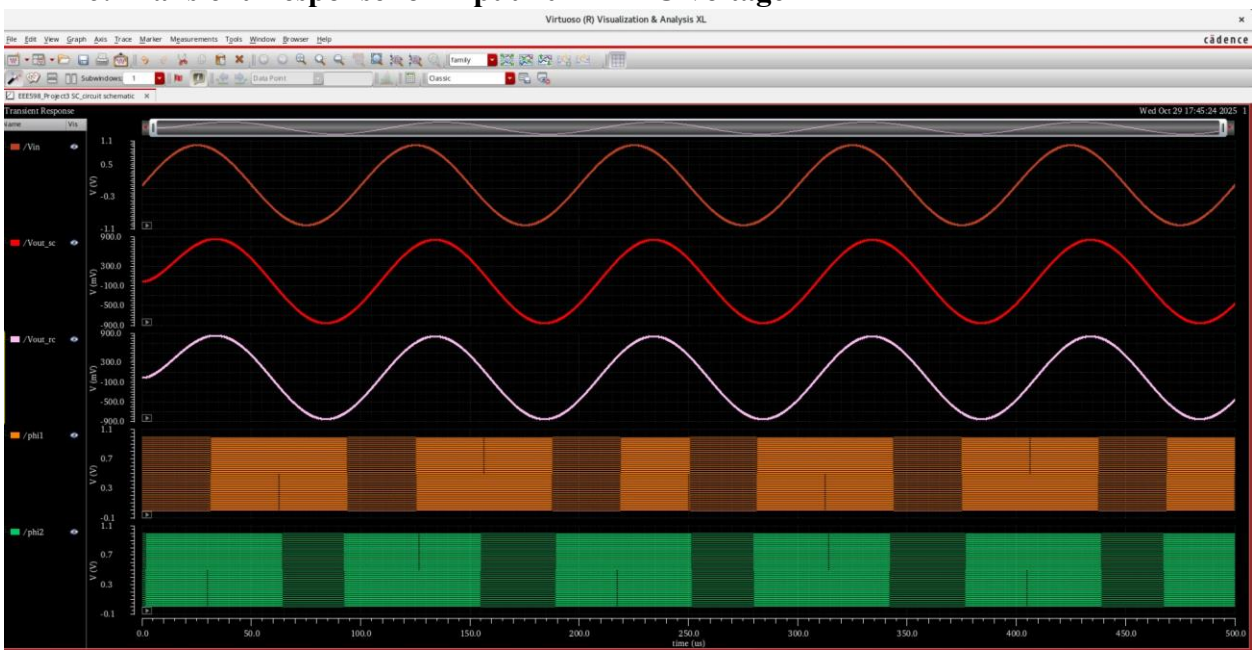


Figure 7: Transient Response for Input 100 kHz AC Voltage

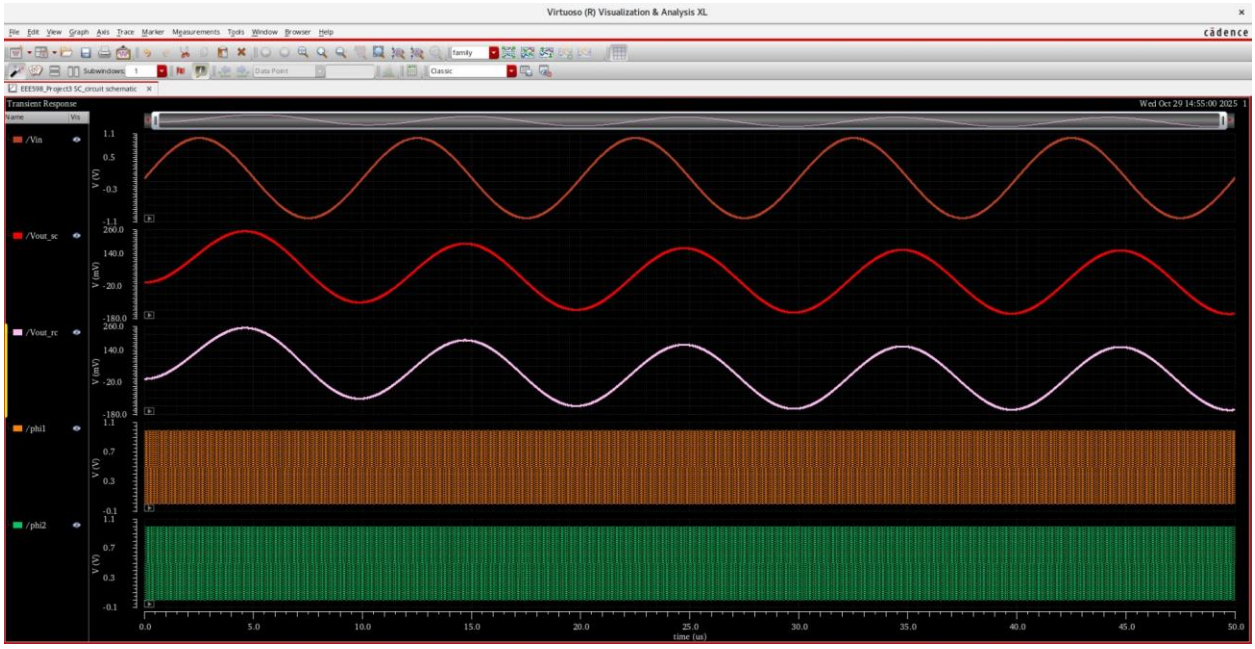


Figure 8 shows the transient response for the 1 MHz AC voltage input. Compared to the other transient responses, these output signals have a far smaller amplitude and experience more delay in comparison to the input signal than the previous steady state responses. As the input AC frequency increases, the phase lag increases as the argument for the phase grows as shown by the offset appearance of the output signals. The much smaller amplitude of the output signals is attributed to the high amount of attenuation of the output signal due to the magnitude of the low-pass-filter falling in the high-frequency range.

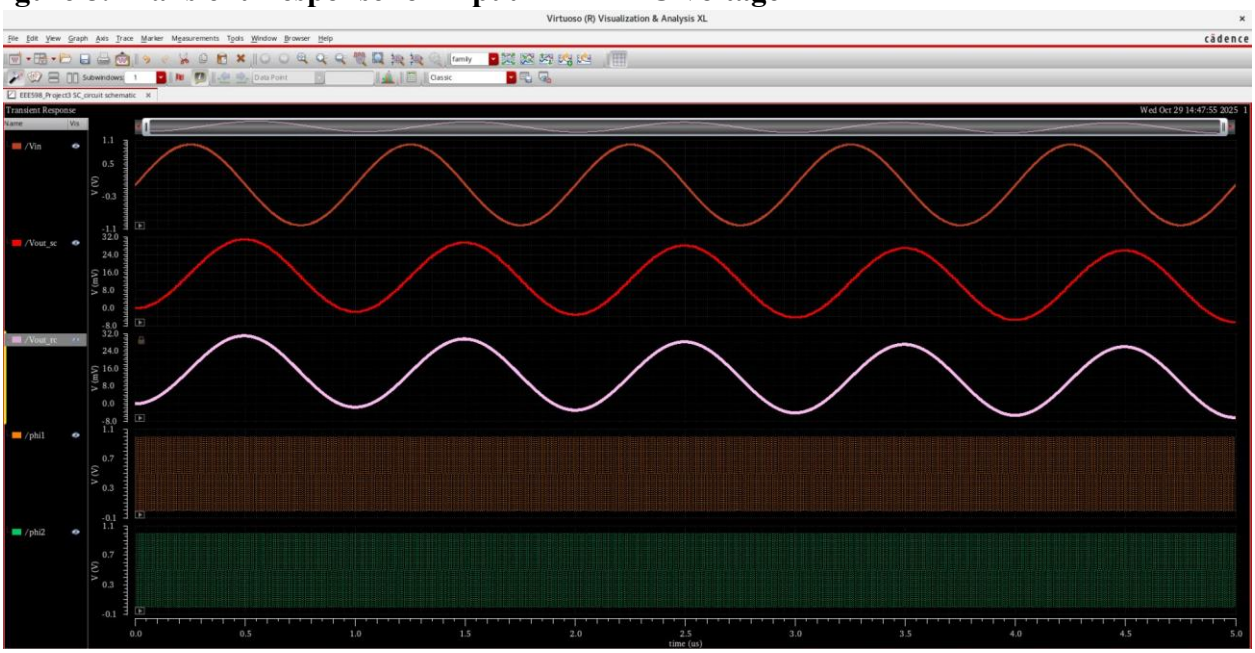



Figure 8: Transient Response for Input 1 MHz AC Voltage